

Changes in the metabolic activities of two agricultural soils as influenced by the pesticides and insecticides combination



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ABSTRACT

Application of pesticides (thiram, difenoconazole, deltamethrin and profenofos) and insecticide combinations (profenofos + cypermethrin and deltamethrin + endosulfan) at concentrations ranging from 0.0 to 10.0 kg ha⁻¹ were tested for their non-target effects towards the activities of protease, invertase, sulphur oxidation, and nitrogen mineralization in two agricultural soils of groundnut under laboratory conditions. In this study, protease activity was measured in terms of tyrosine equivalents formed from casein, and the invertase activity was measured in terms of glucose formed from sucrose. We found pronounced mineralization of peptone-nitrogen and sulphur oxidation in the soil samples treated with 2.5–5.0 kg ha⁻¹ of the insecticide combinations and individual pesticides. Soil samples treated with 2.5 kg ha⁻¹–5.0 kg ha⁻¹ of the insecticide combinations and individual pesticides have shown increased invertase and protease activity at 20th day and then decreased with increasing period of incubation. From the obtained results it was concluded that the influence of these insecticide combinations and individuals on enzyme activity, nitrogen mineralization, and sulphur oxidations were dose dependent.

1. Introduction

In order to meet the demands of a growing population, the use of pesticides has become indispensable in modern agriculture, and that use ultimately influences soil fertility and soil microbiota and their metabolic activities (Cerny et al., 2008; Chu et al., 2008; Munees et al., 2012). In addition, the effect of various combinations of pesticides may deviate from the behaviour of an individual pesticide due to the occurrence of synergistic, antagonistic or additive interactions between or among different pesticides (Schuster and Schroder, 1990). Therefore, in order to determine the ecological significance of pesticides, it has become necessary to study the effects of pesticide combinations applied at recommended levels (Reddy et al., 2011; Volodymyr et al., 2016). Concerns about the side effects of pesticides on soil fertility (Mousumi et al., 2014) led to the development of soil microbial testing programmes (Maleeka Begum and Rajesh, 2015). The testing programme includes measurement of soil enzyme activities which act as biological catalysts for various important biochemical reactions that maintain soil fertility (Ayansina and Oso, 2006). Generally, soil fertility is a reflection of the microbial status of soil, and the pesticides which settle in soil may interrelate with non-target microbiota and mediate significant

biochemical transformations linked with nutrient recycling (Lakshmi et al., 2015; Maleeka Begum and Rajesh, 2015). Any disorder in the soil micro organisms may affect the primary function of the soil which is related to a continually cycled pool of nutrients for plant growth (Munees et al., 2012). Recycling of nutrients in soil organic matter involves the participation of soil microorganisms (Lakshmi et al., 2015) and pesticide application may adversely influence the nitrogen cycle, viz, ammonification, nitrification, denitrification and nitrogen fixation, which in turn may have serious effects on crop production (Pranita et al., 2015). Therefore, greater attention has been paid to study the influence of pesticides on a biological transformation of soils (Gundi et al., 2005; Jacobsen and Hjelmsø, 2014). In view of plant nutrition, sulphur stands next to nitrogen and phosphorus which enters the soil primarily in the form of plant residues, chemical fertilisers etc. In general, the sulphur transformation is microbially mediated by chemolithotrophs and has become increasingly apparent, since sulphate is the plant available source of sulphur (Wen et al., 2001; Srinivasulu et al., 2015). Despite the economic importance of sulphur, knowledge about pesticidal influence towards these microbial activities is scanty and therefore, the present investigation reports the interaction effects of pesticides viz., profenofos, deltamethrin, difenoconazole, thiram and

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insecticide combinations involving cypermethrin + profenofos and deltamethrin + endosulfan on nitrogen mineralization, sulphur oxidation, protease and invertase activities in two agricultural soils.

2. Materials and methods

2.1. Soils

Two agricultural samples of black clay (13°57'45.35" N and 77°50'13.15" E) soil and red sandy loam (13°57'34.54" N and 77°49'56.12" E) soils were collected at a depth of 12 cm random sampling strategy of Anantapur district, a semi-arid zone in Andhra Pradesh, India. Later they were air-dried and sieved through a 2 mm mesh for further assessment.

2.2. Pesticides

Two fungicides, thiram (75% WP) (carbamate), difenoconazole (25% EC) (triazole), and two insecticides, deltamethrin (2.8% EC) (synthetic pyrethroid) and profenofos (50% EC) (organophosphate) and insecticide combinations like profenofos (organophosphate) + cypermethrin (25% EC) (synthetic pyrethroid) and deltamethrin (synthetic pyrethroid) + endosulfan (35% EC) (organochlorine) were used in the present study. Commercial formulations dissolved in distilled water of these tested pesticides were used for determining the microbial activities like soil enzymes, nitrogen mineralization, sulphur oxidation.

2.3. Soil incubation

The soil ecosystem stimulating non-flooded conditions consisting of ten gram of soil samples were added in test tubes (25 × 150 mm) and moistened to a water potential of 0.090 MPa, in order to maintain at 60% water holding capacity (Rangaswamy and Venkateswarlu, 2000).

2.4. Protease activity

Two and the five-gram portion of each soil, in duplicates, were treated with the selected pesticides at 1, 2.5, 5.0, 7.5 and 10 kg ha⁻¹ concentration. Soil samples without pesticide treatment served as control. Soil samples in test tubes with and without pesticide treatment were incubated at room temperature in the lab (28 ± 4 °C). After 10 days of incubation, the soil extract was prepared in distilled water for an assay of proteases (Rangaswamy and Venkateswarlu, 1996).

2.5. Assay of soil protease

Soil samples including controls were incubated with 10 ml of 0.1 M Tris (2-amino-2 (hydroxy-methyl) propane-1:3-diol) (pH-7.5) containing sodium caseinate (2%w/v) for 24 h at 30 °C. An aqueous solution of trichloro acetic acid (4 ml, 17.5% w/w) was then added and the mixer was centrifuged. The supernatant liquid, in suitable aliquots, was treated with 3 ml of 1.4 M Na₂CO₃ and 1 ml of Folin-Ciocalteu reagent (33.3% v/v) with rapid swirling. The blue colour, thus formed after 30 min, was read at 700 nm in a spectronic – 20 D Spectrophotometer (Milton and Roy). Tyrosine equivalents in soil extracts were estimated by referring to a calibration curve prepared with a known concentration of tyrosine.

2.6. Assay of soil invertase

The method employed for assay of invertase was developed by Cole (1997) and followed by Tu et al., (1981). The soil samples were transferred to 100 ml Erlenmeyer flasks and were treated with 1 ml of toluene to arrest the enzyme activity. After 15 min, 6 ml of 18 mM sucrose was added to the soil samples and incubated for 24 and 48 h respectively. The testing samples were then passed through Whatman

No.1 filter paper, and the filtrate was assayed for the amount of glucose by Nelson Somogyi method in spectronic 20 D spectrophotometer. In another experiment, the rate of enzyme activity invertase was determined at 10, 20, 30, 40 days of soil incubation and further with the respective suitable substrate.

2.7. Nitrogen transformation in soil

The influence of selected insecticides at varying concentration on nitrogen mineralization was studied in two agricultural soils. Initially, 0.05 ml aliquots from stock solutions of the insecticides were applied with 0.1 ml pipette to the surface of a 10-g portion of the black and red soils contained in (25 × 200 mm) test tubes. Soil samples receiving only 0.05 ml acetone served as control. After the solvent had evaporated completely at room temperature, all the treatment including control was supplemented with 1000 ppm of nitrogen from AR grade peptone (organic N) for ammonification and 200 ppm nitrogen from (NH₄)₂SO₄ (inorganic N) for nitrification. The soil samples were then homogenised to distribute pesticide and nitrogen source. These soil samples were maintained at 60% water holding capacity throughout the incubation period. After 7 days of incubation, at room temperature (28 ± 4 °C) triplicates of each treatment were withdrawn and extracted with 2 M KCl for analyzing ammonia by Nesslerization method (Jackson, 1971) and with distilled water for estimation of nitrite by the diazotization method (Barnes and Folkard, 1951) and nitrate by the brucine method (Ranney and Barlett, 1972).

2.8. Estimation of ammonia

The ammonia (NH₄⁺ – N) formed in peptone amended soil samples with and without insecticide application was analysed by Nesslerization (Jackson, 1971). To suitable aliquots of the soil extract, 0.5 ml of the Nessler's reagent was added, and the volume was made up to 7 ml. The yellow colour that developed was read at 495 nm in a spectronic – 20 D spectrophotometer (Milton Roy).

2.9. Estimation of nitrite

Nitrite was estimated by diazotization following the method of Barnes and Folkard (1951). Suitable aliquots from the filtrate of the soil extract were pipetted into test tubes and 1 ml of 1% sulphanilamide in 1 N HCl was added and shaken thoroughly. To the coloured diazo compound so formed was added 1 ml of 0.12%, N-(1-naphthyl)-ethylene diaminedihydrochloride in distilled water. The colour was allowed to develop for 25 min, and later the volume was made up to 25 ml with distilled water. The absorbance of the pink coloured solution was read at 520 nm in a Spectronic – 20 D Spectrophotometer. The amount of nitrite was calculated by referring to a calibration curve prepared with the standard solution of known nitrite concentrations.

2.10. Estimation of nitrate

The nitrate (NO₃⁻ – N) produced due to the activity of soil microorganisms was determined by the method of Ranney and Bartlett (1972). Three drops of brucine reagent (2 g brucine dissolved in 50 ml methanol) were added to suitable aliquots of the soil extract followed by 2 ml of concentrated sulphuric acid. The solution was mixed by vortexing and placed in the dark for 30 min to ensure full-colour development, after which the volume was made up to 10 ml with distilled water, and the yellow colour was read at 410 nm in a spectronic – 20 D spectrophotometer. Further, the experiment was repeated with only 2.5 kg ha⁻¹ of the selected insecticides and incubated for 1, 2, 3 and 4 weeks in order to determine the rate of ammonification. The impact of different concentrations of the insecticides after two weeks of the incubation period and the rate of nitrification at 2, 4, 6 and 8 weeks was determined following the same procedure as with ammonification.

2.11. Sulphur oxidation

To determine the effect of selected insecticides on sulphur oxidation, a selection of ten-gram portions of soil taken in 25×150 mm test tubes was supplemented with 25 mg of AR grade sulphur powder. After homogenization, 0.05 ml stock solutions of the insecticides were added to provide final concentrations of 1, 2.5, 5, 7.5 and 10 kg ha^{-1} active ingredient. Soils containing only acetone served as controls. After complete evaporation of the solvent, at room temperature and uniform mixing, samples were incubated at room temperature (in darkness) at $28 \pm 4^\circ\text{C}$ and maintained at 60% water holding capacity. Sulphate sulphur that formed was extracted from triplicate soil samples after seven days and analysed-turbidometrically (Massoumi and Cornfield, 1963). Further, the experiment was repeated with only 2.5 and 5 kg ha^{-1} level of the respective insecticides to determine the rate of Sulphur oxidation at 1, 2, 3, 6 and 12 weeks of incubation.

2.12. Estimation of sulphate

Ten-gram soil samples were suspended in 100 ml sodium acetate-acetic acid buffer solution, thoroughly agitated in a wrist action shaker for 30 min, and the soil suspensions were passed through Whatman filter paper No.42. Suitable aliquots of the filtrate were pipetted into 25 ml volumetric flasks. Then 2.5 ml of 25 HNO_3 and 20 ml acetic phosphoric acid (3 parts of AR grade acetic acid + 1 part of AR grade H_3PO_4) mixture was added to each flask and diluted to about 22 ml. After shaking the contents thoroughly, 0.5 ml of BaSO_4 seed suspension (18 g AR BaCl_2) was dissolved in 44 ml of hot distilled water and 0.5 ml of standard sulphate solution (2 mg S for 1 ml) was added, the solution was then boiled and cooled, and 4 ml of a gum acacia-acetic acid solution was added. Then 0.2 g AR grade BaCl_2 crystals (passed through a 1 mm sieve) were added and mixed thoroughly. After 10 min, 1 ml of gum acacia- an acetic acid solution was added (5 gm of gum acacia was dissolved in 500 ml hot distilled water) and, filtered through Whatman filter paper no.42. The filtrate was cooled, and its volume was made to one litre (with AR grade acetic acid) and set aside for 90 min. The flasks were then inverted ten times, and the absorbance was read in a Spectronic 20-D spectrophotometer at 440 nm.

2.13. Statistical analysis

The concentration of the protease, invertase, sulphur oxidation and nitrogen mineralisation was calculated on the basis of soil weight (oven-dried). Data were analysed using one-way ANOVA, and the differences were contrasted using Duncan's multiple range test (DMRT) (Megharaj et al., 1999; Jaya Madhuri and Rangaswamy, 2009). All the statistical analysis was performed at $P \leq 0.05$ using SPSS statistical software package (Table 1).

3. Results and discussion

3.1. Ammonification

Microbial processes, in particular, associated with the nitrogen cycle, are often chosen as test systems owing to the ecological significance of nitrogen transformation in the biosphere. Hence, ammonification, a first step in the nitrogen mineralisation, was studied under the influence of selected pesticides profenofos, deltamethrin, thiram and difenoconazole and two insecticide combinations profenofos + cypermethrin and deltamethrin + endosulphan in the laboratory condition. The sum of different forms of inorganic nitrogen (NH_4^+ -N, NO_2^- -N and NO_3^- -N) formed from organic nitrogen in soils represents total ammonification activity because NO_2^- -N and NO_3^- -N cannot be made from organic nitrogen without the formation of NH_4^+ -N. The rate of mineralization of peptone - N i.e., ammonification increased significantly by the application of profenofos,

Table 1

Influence of selected pesticides on ammonification in black and red soils.

Name of Pesticide	Soil incubation, in weeks			
	1	2	3	4
BLACK SOIL				
Control (0.0)	420 ^a	480 ^a	450 ^a	387 ^a
Profenofos (2.5 kg ha^{-1})	510 ^b	550 ^b	478 ^b	425 ^b
Deltamethrin (2.5 kg ha^{-1})	492 ^c	525 ^c	480 ^c	410 ^c
Thiram (2.5 kg ha^{-1})	486 ^c	515 ^c	475 ^b	400 ^c
Difenoconazole (2.5 kg ha^{-1})	497 ^c	535 ^d	487 ^c	450 ^b
RED SOIL				
Control (0.0)	382 ^a	450 ^a	360 ^a	345 ^a
Profenofos (5.0 kg ha^{-1})	481 ^b	510 ^b	450 ^b	410 ^b
Deltamethrin (5.0 kg ha^{-1})	495 ^b	525 ^b	456 ^b	405 ^b
Thiram (2.5 kg ha^{-1})	455 ^c	498 ^c	435 ^c	395 ^c
Difenoconazole (2.5 kg ha^{-1})	467 ^c	502 ^c	425 ^c	412 ^b

Values are μg of ($\text{NH}_4^+ + \text{NO}_2^- + \text{NO}_3^-$) - N g^{-1} Soil.

Means, in each column, followed by the same letter are not significantly different ($P \leq 0.05$) from each other according to DMR test.

deltamethrin, thiram and difenoconazole and two insecticide combinations profenofos + cypermethrin and deltamethrin + endosulphan in both soils (Fig. 1 and 2). During 7 days of incubation, profenofos, deltamethrin, thiram and difenoconazole have shown high amounts of mineralization of organic nitrogen at 2.5 kg ha^{-1} (Fig. 1) in black soil, whereas in red soil profenofos and deltamethrin have shown a high amount of mineralization of organic nitrogen at 5.0 kg ha^{-1} . However, in the case of thiram and difenoconazole, a high quantity of mineralization of organic nitrogen appeared at 2.5 kg ha^{-1} (Fig. 2). The highest amount of mineralization of organic nitrogen occurred at 14 day incubation in all tested individual pesticides in both red and black soils, whereas, in the case of insecticide combinations, all tested combinations i.e., profenofos + cypermethrin and deltamethrin + endosulphan, showed a high amount of mineralization of organic nitrogen at 5.0 kg ha^{-1} in both black and red soils (Fig. 3). Similarly, with individual pesticides in insecticide combinations also, the most amount of mineralization of organic nitrogen occurred at 14-day incubation in both red and black soils (Table 2). The mineralization of organic nitrogen was more pronounced especially at the highest concentration of 5.0 kg ha^{-1} in all the tested pesticide and insecticide combinations which reflect field application rates of the pesticides. There was a rise in the formation of inorganic-N in both soils with/without pesticides at the 14-day interval, followed by a slight decline at the 28-day interval. In the present study, all the tested pesticide and insecticide combinations were inoffensive or non-toxic to ammonification in both soils. The interaction responses such as synergistic, additive and antagonistic interactions occurred toward the ammonification process when the insecticides were applied to both soils in combinations. Correspondingly, monocrotophos and quinalphos up to 2.5 kg ha^{-1} and cypermethrin and fenvalerate at 5 kg ha^{-1} considerably stimulated the ammonification but higher concentrations of these insecticides were toxic to the mineralization process in two agricultural soils (Rangaswamy and Venkateswarlu, 1990,1993). Srinivasulu et al. (2012) revealed that the chlorpyrifos and monocrotophos single and combination with carbendazim and mancozeb significantly increased the population of *Azospirillum* spp and ammonification in groundnut soils at 1.0 – 5.0 kg ha^{-1} . The study performed by Walia et al. (2014) revealed that the decreased ammonification at 10 ppm of mancozeb in apple orchard soil. Stefania and Agata (2015) demonstrated that the herbicides reglone 200SL and Elastiq 550 EC were applied to the soil at recommended doses did not cause any significant disturbance processes of ammonification and nitrification. Tu (1989) reported that ammonium production was not appreciably altered by the treatments of DOWCO 429 $\times$, DPX 43898, tefluthrin and trichloronate in a sandy loam and an organic soil. These results clearly indicate the additive

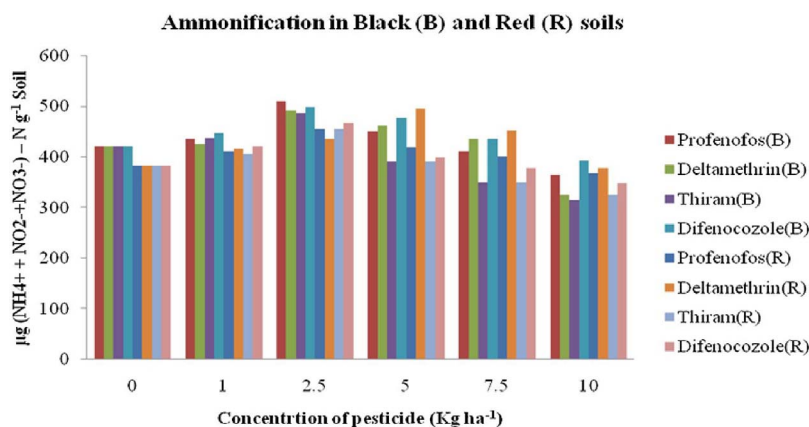


Fig. 1. Effect of different concentrations of selected pesticides on ammonification in black and red soil after 7 days.

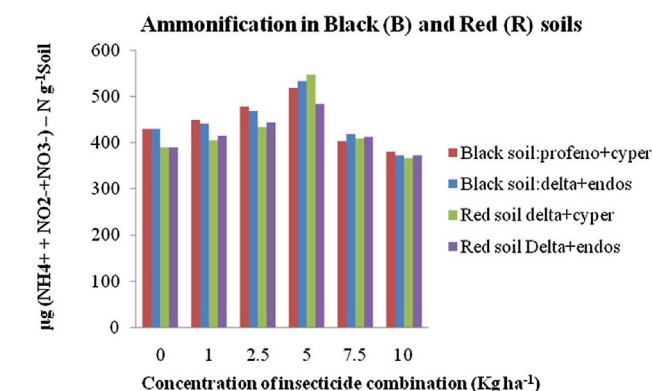


Fig. 2. Effect of different concentrations of selected insecticide combinations on ammonification in red soil and black soil after 7 days.

interaction between insecticides in the soil. The overall effects of combinations of pesticides on microbial activities in soil may be subjected to interactions between pesticides, such as additive, synergistic and antagonistic effects, and may deviate from the behaviour of the individual pesticide components (Auspurg, 1985). These results of the present study clearly suggest that insecticide combinations, profenofos + cypermethrin and deltamethrin + endosulphan, could interact to alter their influence considerably on non-target microbiota and their activities in soil and modify the environmental impact of the microbiota.

3.2. Nitrification

Nitrification is perhaps the most commonly used soil microbial process for assessing effects of pesticides. Autotrophic nitrification is

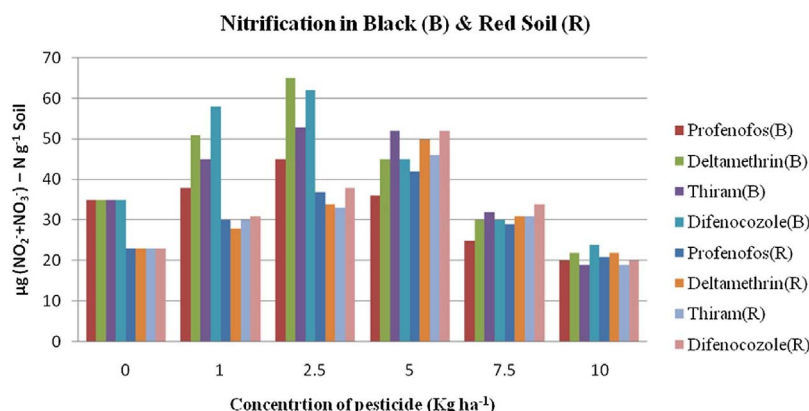


Table 2

Effect of selected insecticide combinations at 5.0 kg ha⁻¹ on ammonification in black and red soils.

Treatment	Soil incubation, in weeks			
	1	2	3	4
BLACK SOIL				
Control (0.0)	430 ^a	490 ^a	450 ^a	410 ^a
Profenofos + Cypermethrin (5.0 kg ha ⁻¹)	520 ^b	570 ^b	480 ^b	435 ^b
Deltamethrin + Endosulphan (5.0 kg ha ⁻¹)	535 ^b	567 ^b	485 ^c	420 ^b
RED SOIL				
Control (0.0)	390 ^a	410 ^a	350 ^a	340 ^a
Profenofos + Cypermethrin (5.0 kg ha ⁻¹)	490 ^b	505 ^b	425 ^b	400 ^b
Deltamethrin + Endosulphan (5.0 kg ha ⁻¹)	484 ^b	510 ^b	454 ^b	394 ^b

Values are μg of (NH₄⁺ + NO₂⁻ + NO₃⁻) - N g⁻¹ Soil.

Means, in each column, followed by the same letter are not significantly different ($P \leq 0.05$) from each other according to DMR test.

carried out in two steps: oxidation of NH₄⁺ to NO₂⁻ which, in turn, further is oxidized to NO₃⁻. Thus, nitrification transforms less mobile soil nitrogen into a mobile form. Hence, nitrification in two soils under the impact of insecticides in combinations was studied under laboratory condition. The sum of NO₂⁻ - N and NO₃⁻ - N, formed from NH₄⁺ - N in the soil, represents total nitrifying activity. The rate of nitrification under the influence of selected pesticides profenofos, deltamethrin, thiram and difenoconazole progressively increased with 2.5 kg ha⁻¹ in black soil and afterwards increasing the concentration of pesticides decreased the rate of nitrification and decline phase was started. However, in the case of red soil, the rate of nitrification was increased up to 5.0 kg ha⁻¹ and afterwards, the decline phase was started at 14-day incubation (Fig. 3). Utmost nitrification occurred after 4 weeks of incubation in all tested individual pesticides in both red and black soils (Table 3). With insecticide combinations, all case of

Fig. 3. Effect of different concentrations of selected pesticides on nitrification activity in black and red soil after 14 days.

Table 3
Influence of selected pesticides on nitrification activity in black and red soils.

Name of Pesticide	Soil incubation, in weeks			
	2	4	6	8
BLACK SOIL				
Control (0.0)	35 ^a	110 ^a	80 ^a	51 ^a
Profenofos (2.5 kg ha ⁻¹)	45 ^b	123 ^b	93 ^b	63 ^b
Deltamethrin (2.5 kg ha ⁻¹)	65 ^c	133 ^c	101 ^c	75 ^c
Thiram (2.5 kg ha ⁻¹)	53 ^d	125 ^b	97 ^b	52 ^a
Difeneconazole (2.5 kg ha ⁻¹)	62 ^c	145 ^d	111 ^d	71 ^a
RED SOIL				
Control (0.0)	23 ^a	88 ^a	73 ^a	45 ^a
Profenofos (5.0 kg ha ⁻¹)	42 ^b	101 ^b	66 ^b	53 ^b
Deltamethrin (5.0 kg ha ⁻¹)	50 ^c	112 ^c	73 ^c	51 ^b
Thiram (5.0 kg ha ⁻¹)	46 ^b	93 ^b	67 ^b	41 ^c
Difeneconazole (5.0 kg ha ⁻¹)	52 ^c	105 ^d	81 ^c	54 ^b

Values are μg of ($\text{NO}_2^- + \text{NO}_3^-$) – N g^{-1} Soil.

Means, in each column, followed by the same later are not significantly different ($P \leq 0.05$) from each other according to DMR test.

insecticide combinations all tested combinations i.e., profenofos + cypermethrin and deltamethrin + endosulphan showed a high level of nitrification at 2.5 kg ha⁻¹ after 14 days incubation in black soil, but red soil 5.0 kg ha⁻¹ showed the maximum nitrification (Fig. 4). As in the case of individuals, with insecticide combinations, utmost nitrification occurred after 4 weeks of incubation in all tested insecticide combinations in both red and black soils (Table S1). The experimental data showed that the oxidation of NH_4^+ to NO_3^- through NO_2^- decreased after 5.0 kg ha⁻¹ in both black and red soils, and with all the four pesticides and insecticide combinations used in this study. It is confirmed that these pesticides were not inhibitory to the nitrification at concentrations up to 2.5 to 5.0 kg ha⁻¹ in both soils. There was the greater accumulation of $\text{NO}_2^- + \text{NO}_3^-$ in soils treated with pesticides and insecticide and combinations at the end of a 4-week incubation. In

another study of Rangaswamy and Venkateswarlu (1990, 1993), cypermethrin, monocrotophos, quinalphos, and fenvalerate at concentrations ranging from 1.0 to 5.0 kg ha⁻¹ significantly stimulated the nitrification, especially at 2.5 kg ha⁻¹. This observation was in conformity with the results of the present study. Srinivasulu et al. (2012) suggested that the pesticides chlorpyrifos and monocrotophos alone in combination with carbendazim and mancozeb enhanced the nitrification at 2.5–5.0 kg ha⁻¹ in two agricultural soils. The study suggested by Man and Zucong (2009) revealed that the activity nitrification inhibited at the rate of 5.5 mg kg⁻¹ chlorothalonil in two agricultural soils. The application of mancozeb caused significantly decrease on nitrification process to 11.2%–5.6% in arable and grassland soils (Černohlávková et al., 2009). Similarly, the lethal effects of mancozeb on nitrification was studied by other authors (Kinney et al., 2005; Cycon et al., 2010). In turn, the nitrification process was significantly increased at 0, 10, 100, 250, and 500 ppm mancozeb at the end of one week of incubation in soils (Walia et al., 2014). Nitrification was also affected by the combination of insecticides in both soils in the same way as observed with the ammonification in the soil samples of the present study (Supplementary Table 1). These results clearly indicate an antagonistic interaction between insecticide combinations in both soils. Similarly, Dichlorprop, glyphosate (herbicides), Sportak-alpha, Bayleton DF, Corbel (fungicides) and pirimicarb (insecticide) in different combinations at field rates and 10-fold field rates inhibited nitrification in a laboratory while the same pesticides stimulated the nitrification in the field (Schuster and Schroder, 1990). On the contrary, 19 years of cumulative annual field applications of benomyl, chlorfenvinphos, aldicarb, triadimefon and glyphosate either singly or in combinations at the recommended field rates, had no influence on the mineralization of organic-N to nitrate through ammonium, without measurable long-term harmful effects in spite of continuous usage of these pesticides (Hart and Brookes, 1996). Similarly, monocrotophos, quinalphos, cypermethrin and fenvalerate individually have shown the significant stimulatory effect on nitrification, at concentrations ranging from 1.0 to 5.0 kg ha⁻¹ (Gundi Vijay et al., 2006). As opposed to these

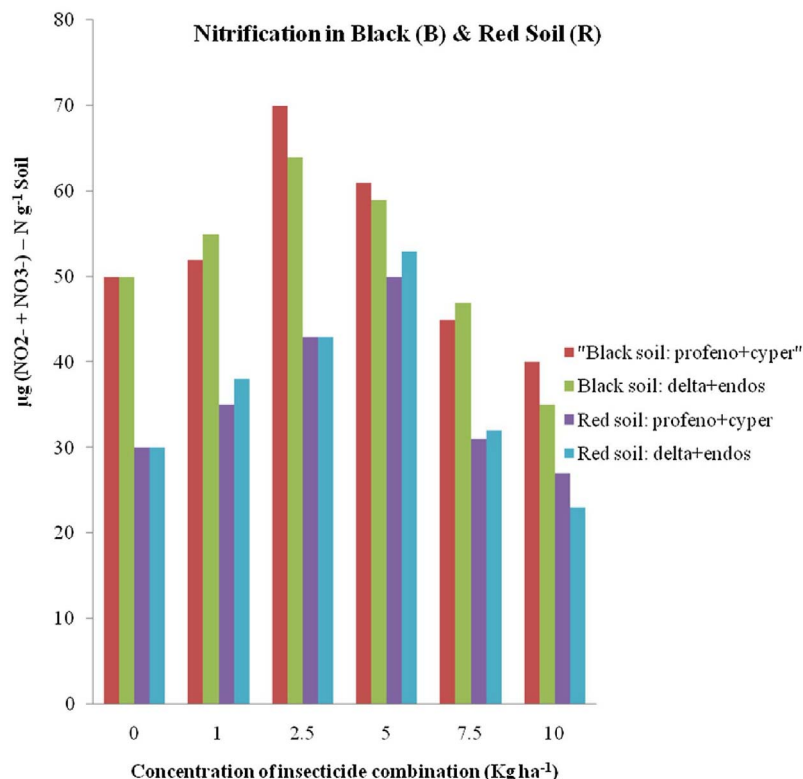


Fig. 4. Effect of different concentrations of selected insecticide combinations on nitrification activity in black and red soils after 14 days.

observations, Caceres et al. (2009) determined that organophosphate insecticide fenamiphos even at 10 mg kg⁻¹ soil (1.0 kg ha⁻¹) dramatically reduced the nitrification.

3.3. Sulphur oxidation

The effect of selected pesticides (profenofos, deltamethrin, thiram and difenoconazole) and two insecticide combinations (profenofos + cypermethrin and deltamethrin + endosulphan) were tested for the oxidation of sulphur in terms of sulphate formed. Oxidation of elemental sulphur in terms of sulphate formed proceeded in black soil treated with the profenofos, deltamethrin, thiram up to 2.5 kg ha⁻¹ but in the case of difenoconazole, it was 5.0 kg ha⁻¹. In the case of red soil oxidation of elemental sulphur occurred with 2.5 kg ha⁻¹ in profenofos and deltamethrin but 5.0 kg ha⁻¹ in thiram and difenoconazole after a 7-day incubation (Supplementary Fig. S1). Application of profenofos, deltamethrin and thiram at 2.5 kg ha⁻¹ and difenoconazole at 5.0 kg ha⁻¹ progressively increased the sulphur-oxidizing activity of the black soil up to an 84 day incubation, whereas application of profenofos and deltamethrin at 2.5 kg ha⁻¹ and Thiram and difenoconazole at 5.0 kg ha⁻¹ progressively increased the sulphur-oxidizing activity of the red soil up to an 84 day incubation (Table S2). Oxidation of elemental sulphur in terms of sulphate formed proceeded in both black and red soils treated with the insecticide combinations profenofos + cypermethrin and deltamethrin + endosulfan up to 2.5 kg ha⁻¹ after a 7-day incubation (Fig. 5). Application of insecticide combinations profenofos + cypermethrin and deltamethrin + endosulfan at 2.5 kg ha⁻¹ progressively increased the sulphur-oxidizing activity of the black and red soils up to an 84-day incubation (Table 4). A similar increase in the rate of sulphur oxidation has been reported by several workers after the addition of a variety of compounds. Sivasithamparam, (1969) found that chlorpyrifos treated soil showed a distinct increase in the number of sulphate oxidizers. This rise in number was more pronounced three months after submergence. Benomyl when applied at 1 and 250 mg g⁻¹ to soil often resulted in an increase in SO₄²⁻-S. Likewise, the addition of commercial formulation of benomyl in alluvial and laterite soils was stimulatory at 5 and 10 µg g⁻¹, in flooded soil (Ray et al., 1980). Tu (1994) also established that fungicides, dodine, chlorothalonil, folpet, thiram and captan increased Sulphur oxidation in loamy sand soil under lab conditions. The population of sulphur oxidising strains increased in the presence of malathion and dimethoate (Bezbaruah et al., 1995). Further, Tu (1995) noticed that five insecticides, clyfluthorin, tebuperimphos, tebupiriphos, amitraz and imidacloprid stimulated sulphur oxidation after 4 weeks in sandy soil. Similar evidence was provided by Tu (1996) with herbicides in sandy loam soil. In a similar manner, application of monocrotophos, quinalphos, cypermethrin and fenvalerate promoted sulphur oxidation up to 7.5 kg ha⁻¹ in groundnut soils (Rangaswamy and Venkateswarlu, 1999). This kind of stimulation was also found by Min et al. (2001) on the application of butachlor. Interestingly, a level of 7.5 kg ha⁻¹ of the insecticidal treatment which was mostly inhibitory to soil enzymatic

Table 4

Influence of Insecticide combinations on Sulphur oxidation in black and red soils.

Treatment	Soil incubation, in days				
	7	14	21	42	84
BLACK SOIL					
Control (0.0)	700 ^a	810 ^a	905 ^a	1025 ^a	1152 ^a
Profenofos + Cypermethrin(2.5 kg ha ⁻¹)	780 ^b	920 ^b	1051 ^b	1103 ^b	1205 ^b
Deltamethrin + Endosulphan (2.5 kg ha ⁻¹)	800 ^c	950 ^b	1105 ^b	1235 ^c	1375 ^c
RED SOIL					
Control (0.0)	620 ^a	700 ^a	810 ^a	980 ^a	1002 ^a
Profenofos + Cypermethrin (2.5 kg ha ⁻¹)	700 ^b	815 ^b	925 ^b	1020 ^b	1155 ^b
Deltamethrin + Endosulphan (2.5 kg ha ⁻¹)	710 ^b	802 ^b	970 ^c	1150 ^c	1215 ^c

Values are µg of SO₄²⁻ S g⁻¹ soil formed after added sulphur of 2.5 mg g⁻¹ soil.

Means, in each column, followed by the same letter are not significantly different (P ≤ 0.05) from each other according to DMR test.

activities and microbial population (Jaya Madhuri and Rangaswamy, 2002) had no influence on sulphur transformation with dichlorvos, phorate and methomyl in soil samples, whereas a stimulating effect was noticed with respect to chlorpyrifos and methyl parathion in black soil (Jaya Madhuri and Rangaswamy, 2003). In a similar manner, carbofuran, chlorpyrifos, diazinon, ethoprop, fensulfothion, thionazin and tri chlorinate had little or no significant deleterious effect on sulphur oxidation in a sandy soil (Tu, 1970, 1972, 1973). Bollen and Tu (1971) stated that endin applied to soil at more than three times the maximum rate exerted no appreciable effect on the oxidation of elemental sulphur. The present results clearly indicate that high concentration of pesticides and insecticide combinations were inhibitory to sulphate formation in two soils (Fig. 6, Supplementary Fig. S2; Table 4, Supplementary Table S2). Ray and Sethunathan (1989) further coded that HCH (5 ppm) and benomyl (10 ppm) inhibited the oxidation of elemental sulphur in unsaturated soil. Similar evidence was suggested by Bezbaruah et al. (1995) with fenitrothion and endosulfan. In all the soils, supplemented with elemental sulphur, there was a progressive increase in the formation of SO₄²⁻ S throughout the incubation period. Thus, the rate of sulphur oxidation in soils treated and untreated increased significantly by the end of 84 days as compared to that of 7 days. In a similar manner, a sandy loam soil and an organic soil treated with pesticide stimulated SO₄²⁻ formation during the entire 8-week incubation (Tu, 1988). Srinivasulu et al. (2015) reported, that the rate of sulphur oxidation was dramatically increased at 2.5 kg ha⁻¹, whereas an increase in the concentration of pesticides at 7.5–10 kg ha⁻¹ level drastically decreased the rate of sulphur oxidation. The sulphur oxidation was pronounced more at 14-days of incubation with pesticides alone and in combination in both black and red soils. The oxidation of sulphur was significantly enhanced by the application of monocrotophos in combination with mancozeb after 7 and 14-days of incubation in black soil. It was observed that the pesticides like endosulfan and dithane M-45 served a detrimental effect on

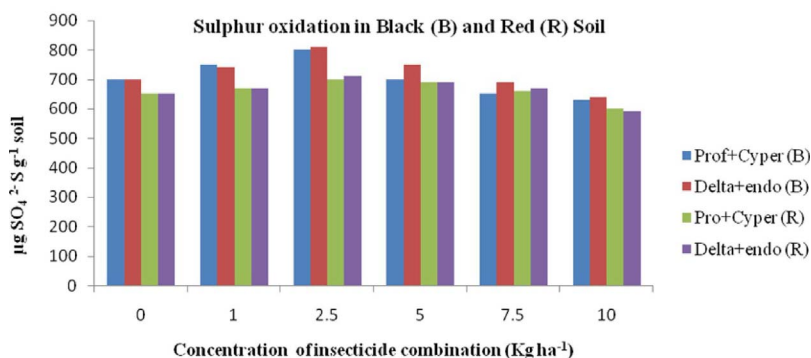


Fig. 5. Effect of different concentrations of selected insecticide combinations on Sulphur oxidation in black and red soils after 7 days.

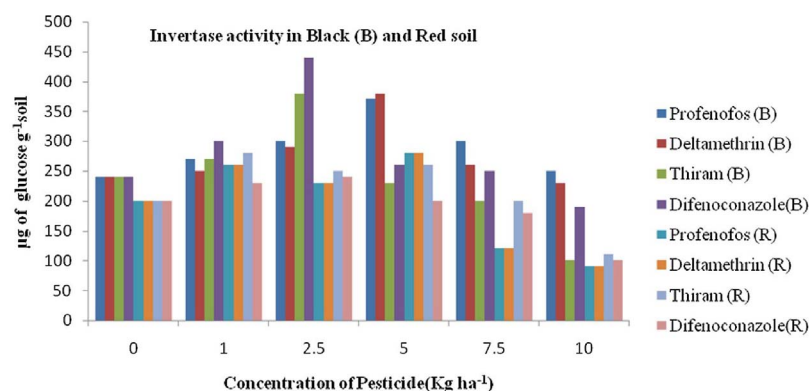


Fig. 6. Effect of selected pesticides on invertase activity in black and red soil after 24 h.

the transformation of S, whereas 2, 4-D created a favourable beneficial effect on S transformation in the soil environment. Pradip et al. (2011) was observed that the pesticides viz., endosulfan and dithane M-45 served a detrimental effect on the transformation of Sulphur, whereas 2, 4-D created a favourable beneficial effect on S transformation in the environment of soil.

3.4. Invertase activity

Invertase enzyme activity was expressed as the amount of glucose formed from the substrate sucrose at 24 and 48 h in soils incubated for 10 days under the influence of selected insecticides. An elevated level of glucose was liberated from the substrate, at the corresponding conditions up to 5 kg ha⁻¹ of both individual pesticide and insecticide combination level. It is also evident that invertase activity was higher in black soil than in red soil due to its high organic matter content (Fig. 6 and 7, Supplementary Fig. S2, S3; Tables 5 and 6 (Supplementary Tables S3, S4). Invertase activity was improved considerably by the application of profenofos and deltamethrin up to 5 kg ha⁻¹ and thiram and difenoconazole up to 2.5 kg ha⁻¹ in black soil, whereas in the case of red soil profenofos and difenoconazole significantly enhanced the amylase activity at 2.5 kg ha⁻¹. In the case of deltamethrin and thiram enhanced activity was observed at 5.0 kg ha⁻¹ after the 24 h incubation for 10 days (Fig. 6). Profenofos and deltamethrin at 5.0 kg ha⁻¹ and thiram and difenoconazole at 2.5 kg ha⁻¹ enhanced the invertase activity in the black soil after 48 h of incubation for 10 days (Supplementary Fig. S2). All the individual pesticides have shown the enhanced invertase activity after 24 h and 48 h of incubation of soil with sucrose and after 20 days of incubation in both black and red soil (Table 5,

Supplementary Table S3). After 20 days, a decline in enzyme activity was observed and lethal activity was shown after 40 days of incubation in both soils with all individual pesticides. In the case of insecticide combinations, profenofos + cypermethrin has shown maximum invertase activity at the concentration of 5.0 kg ha⁻¹ in both black soil and red soil. In the case of deltamethrin + endosulphan, maximum invertase activity was observed at 2.5 kg ha⁻¹ in both red and black soils after 24 h of incubation for 10 days (Fig. 7). In the case of 48 h of incubation for 10 days, the combination of profenofos + cypermethrin has shown maximum invertase activity in the black soil treated with 5.0 kg ha⁻¹ but in the red soil, it was at 2.5 kg ha⁻¹. In the case of the combination of deltamethrin + endosulphan, maximum invertase activity was observed in black soil treated with 2.5 kg ha⁻¹ but in red soil maximum invertase activity was observed at 1.0 kg ha⁻¹. From these results, it was confirmed that the combination of deltamethrin + endosulphan with high concentration, i.e., more than 1.0 kg ha⁻¹, will show a lethal effect on invertase activity (Supplementary Fig. 3). After confirming effective insecticide combination concentration at 24 h and 48 h after treatment, the soils were tested for different (10, 20, 30, 40 days) soil incubation for invertase activity. All the tested insecticide combinations have shown the enhanced invertase activity after 24 and 48 h of incubation of soil with starch and after 20 days of incubation in both black and red soil (Table 6, Supplementary Table S4). After 20 days of incubation, a decline in invertase activity was observed. In present findings, 7.5 and 10 kg ha⁻¹ of all tested pesticides and insecticides, the insecticide concentration showed inhibition of enzyme activity. Palaniappan and Balasubramanian (1985) observed inhibition at 2.5 and 10 ppm of carbofuran on invertase activity in lab conditions. Similarly, El Hamady and Shelo (1999) found that the insecticidal activity of imidacloprid significantly reduced invertase activity in Egyptian soil. Further, Singh et al. (2002) noticed with chlorothalonil, inhibitory effect on enzyme activity at repeated doses in soil samples. Unlike amylase activity, it showed maximum stimulation in soils incubated for 20 days, and further with the substrate at 48 h and exhibited downward trend with increasing period of incubation. The rate of invertase activity followed the same trend of initial stimulation followed by inhibition, as reported by Rangaswamy and Venkateswarlu (1992; Rangaswamy and Venkateswarlu (1992, 1999). On the contrary, Tu (1995) affirmed initial inhibition followed by recovery with five insecticides in sandy loam soil. Ramudu et al. (2011) observed that individual application of the two fungicides at 1.0, 2.5, and 5.0 kg ha⁻¹ to the soil distinctly enhanced the activities of cellulase and invertase but at higher concentrations of 7.5 and 10 kg ha⁻¹ was toxic or innocuous to both cellulase and invertase activities. In soil samples receiving 2.5 – 5.0 kg ha⁻¹ of the fungicides, the accumulation of reducing sugar was pronounced more at 20 days, and the activity of the cellulase and invertase was drastically decreased with an increasing period of incubation up to 30 and 40 days. Yan et al. (2011) noticed that the carbendazim (8.0 mg kg⁻¹) and chloramphenicol showed inhibitory effect on invertase activity. The activity of invertase was

Table 5
Influence of selected pesticides on invertase activity in black and red soil after 48 hours

Name of Pesticide	Soil incubation, in days			
	10 days	20 days	30 days	40 days
BLACK SOIL				
Control (0.0)	330 ^a	490 ^a	434 ^a	300 ^a
Profenofos (5.0 kg ha ⁻¹)	434 ^b	520 ^b	480 ^b	390 ^b
Deltamethrin (5.0 kg ha ⁻¹)	412 ^b	520 ^b	456 ^a	336 ^c
Thiram (2.5 kg ha ⁻¹)	390 ^c	500 ^b	412 ^a	304 ^a
Difenoconazole (2.5 kg ha ⁻¹)	424 ^b	510 ^b	434 ^a	338 ^c
RED SOIL				
Control (0.0)	210 ^a	358 ^a	309 ^a	180 ^a
Profenofos (2.5 kg ha ⁻¹)	390 ^b	430 ^b	350 ^b	246 ^b
Deltamethrin (5.0 kg ha ⁻¹)	320 ^c	480 ^c	370 ^b	267 ^b
Thiram (5.0 kg ha ⁻¹)	390 ^b	468 ^c	350 ^b	280 ^b
Difenoconazole (2.5 kg ha ⁻¹)	356 ^d	420 ^b	324 ^b	304 ^c

µg of glucose g⁻¹ soil formed after 24 h of incubation with sucrose at 28 °C. Means, in each column, followed by same letter or not significantly different ($P \leq 0.05$) from each other according to DMR test.

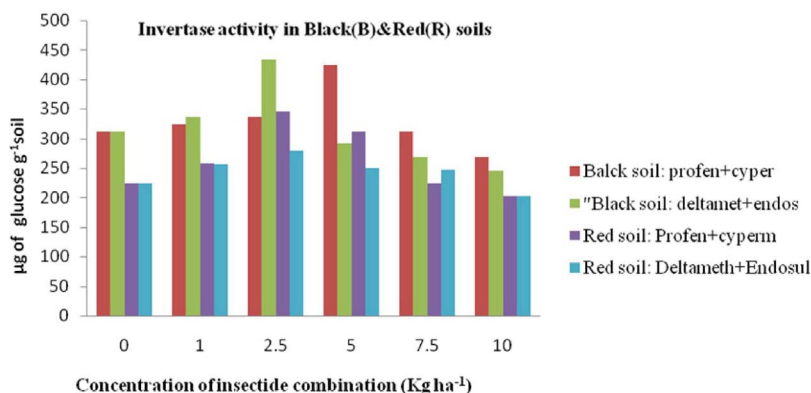


Fig. 7. Effect of selected insecticide combinations on invertase activity in black and red soil after 24 h.

Table 6

Influence of selected insecticide combinations on invertase activity in black and red soils after 48 hours

Name of Pesticide	Soil incubation, in days			
	10 days	20 days	30 days	40 days
BLACK SOIL				
Control (0.0)	346 ^a	520 ^a	424 ^a	280 ^a
Profenofos + Cypermethrin (5.0 kg ha ⁻¹)	424 ^b	570 ^b	530 ^b	336 ^b
Deltamethrin + Endosulphan (2.5 kg ha ⁻¹)	510 ^c	546 ^c	510 ^c	280 ^c
RED SOIL				
Control (0.0)	356 ^a	434 ^a	410 ^a	258 ^a
Profenofos + Cypermethrin (2.5 kg ha ⁻¹)	400 ^b	468 ^b	424 ^b	268 ^b
Deltamethrin + Endosulphan (1.0 kg ha ⁻¹)	500 ^c	534 ^c	490 ^c	246 ^c

µg of tyrosine g⁻¹ soil formed after 24 h incubation with sodium caseinate.

Means, in each column, followed by the same letter are not significantly different ($P \leq 0.05$) from each other according to DMR test.

significantly inhibited by chlorothalonil up to 37.7%, 13.9, and 34.2% respectively Yun et al. (2006). In the study of Vineela Deborah et al. (2013), the results of the experiment using imidacloprid and triadimefon in combination and single on invertase activity in tomato soils showed maximum inhibition was observed at all examined dosages (0.2, 0.5, 0.7 kg ha⁻¹) after 24 h incubation. After 48 h, imidacloprid showed enhanced the invertase activity at 0.5 µg g⁻¹.

3.5. Protease

Proteases promote the breakdown of proteinaceous substances in the soil to simpler nitrogen compounds that are available for plant nutrition. Protease activity was measured in terms of tyrosine equivalents formed from casein in soil samples under the influence of profenofos, deltamethrin, thiram and difenoconazole and the insecticide combinations profenofos + cypermethrin and deltamethrin + endosulfan at different concentrations. In the present study, like other enzyme activities, protease activity of both soils with or without insecticides initially increased up to 20 days and then declined in both soils with/without insecticides (Fig. 8, Supplementary Fig. S4; Tables 7 and 8). Protease activity increased in all individual tested pesticide-treated soils throughout the experiment. The stimulation of the activity of protease increased up to 2.5 kg ha⁻¹ and then declined in the case of profenofos and thiram, whereas in the case of deltamethrin and difenoconazole the increase was at 5.0 kg ha⁻¹ in black soil. However, in red soil, all the tested pesticides have shown enhanced enzyme activity at 2.5 kg ha⁻¹ (Supplementary Fig. S4). This increase in enzyme activity was continued up to 20 days of incubation and afterwards, there was a decline in protease enzyme activity (Table 7). In the case of an insecticide, the combination profenofos + cypermethrin has shown improved protease activity at 2.5 kg ha⁻¹ in both black and red soils.

But with the combination of deltamethrin + endosulphan there was variation in protease activity and unexpectedly high concentration, i.e., 5.0 kg ha⁻¹, has shown maximum enzyme activity with nearly 57% higher over control (Fig. 8) in red soil; in black soil, maximum the enzyme activity was observed at 2.5 kg ha⁻¹. As with individuals, in combinations also enzyme activity was continued up to 20 days of incubation (Table 8). Lethal effects were observed if the incubation period was increased more than 20 days. From the obtained results it was confirmed that protease activity in both soils was stimulated by all four pesticides and both insecticide combinations at all different concentrations used in this study. Similarly, stimulation of protease activity by profenofos, deltamethrin, thiram and difenoconazole used alone was also observed even at later intervals, such as a 20-day interval. Similar observations on protease activity in soil treated with the same insecticides alone were made by Rangaswamy et al. (1994). According to this study, enhancement in protease activity in soils treated with cypermethrin, monocrotophos and quinalphos up to 25 ppm level was recorded. But these insecticides were innocuous or toxic to protease activity in soils at levels higher than 25 ppm. Practically, little information is available on interaction effects of different agrochemicals in combination towards soil protease enzyme that is, involved in nutritional recycling. Protease activity in soils was individually inhibited by application of urea at 200- N kg ha⁻¹ and of herbicides dalapon and paraquat at 10 kg ha⁻¹ by 13 and 18%, respectively. A combination of urea and herbicides at same concentrations reduced protease activity in soils by 41–56% indicating synergistic interaction at the end of a 10-day incubation (Namdeo and Dube, 1973). However, divergent interactions between insecticides on the activity of protease occurred depending on the concentration of interacting insecticides in the present study. These interaction patterns may have the reflective influence on soil enzymes, thereby affecting soil fertility. The persistent interaction responses observed in the present study may not be related to only parent insecticides because of their ephemeral nature in the soil environment (Roberts and Standen, 1981; Lee et al., 1990). In the study of Jastrzebska (2011), the results of the experiment using chlorpyrifos and teflubenzuron indicate that both insecticides modified the biochemical parameters of soil. Jaffer Mohiddin et al. (2013) found that the insecticides, acephate and imidacloprid were profoundly increased the activity of protease and invertase at 1.0 to 2.5 kg ha⁻¹ in agricultural soils. Nazima and Zafar (2010) observed that the application of mancozeb pesticide, particularly N 100 treatment stimulated the protease activity in silt loam soils.

4. Conclusion

The results of the present study reveal that the application of individual pesticide and in combinations at field application rates 1.0 kg ha⁻¹ to 5.0 kg ha⁻¹ enhanced the activities of nitrogen mineralisation, sulphur oxidation, invertase and protease significantly in both black and red soils. However high concentrations of (7.5–10 kg ha⁻¹)

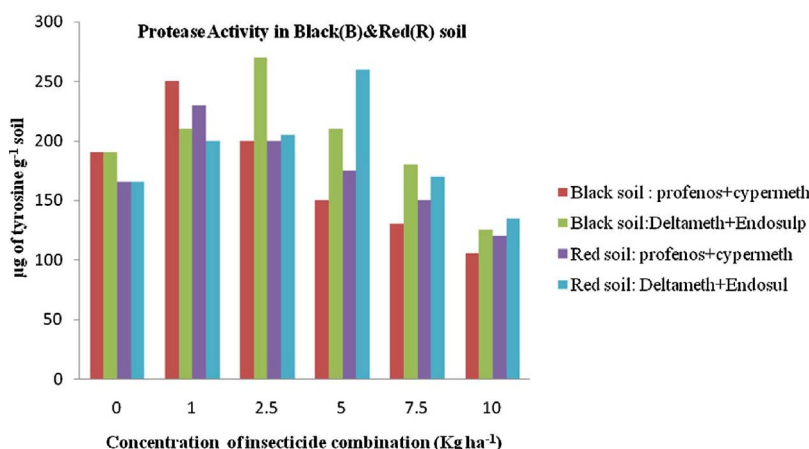


Fig. 8. Effect of different concentrations of the selected pesticides on protease activity in black and red soil after 10 days.

Table 7
Influence of the selected pesticides on protease activity in black and red soils.

Name of Pesticide	Soil incubation, in days			
	10 days	20 days	30 days	40 days
BLACK SOIL				
Control (0.0)	270 ^a	410 ^a	250 ^a	195 ^a
Profenofos (2.5 kg ha ⁻¹)	360 ^b	410 ^a	300 ^b	210 ^a
Deltamethrin (5.0 kg ha ⁻¹)	360 ^b	440 ^b	290 ^b	260 ^c
Thiram (5.0 kg ha ⁻¹)	340 ^b	430 ^a	290 ^b	240 ^b
Difeneconazole (5.0 kg ha ⁻¹)	350 ^b	490 ^c	250 ^a	200 ^a
RED SOIL				
Control (0.0)	190 ^a	210 ^a	160 ^a	140 ^a
Profenofos (2.5 kg ha ⁻¹)	260 ^b	300 ^b	230 ^b	200 ^b
Deltamethrin (2.5 kg ha ⁻¹)	260 ^b	340 ^b	230 ^b	180 ^c
Thiram (2.5 kg ha ⁻¹)	260 ^b	320 ^b	250 ^c	200 ^b
Difeneconazole (2.5 kg ha ⁻¹)	250 ^c	300 ^b	220 ^b	190 ^c

µg of tyrosine g⁻¹ soil formed after 24 h incubation with sodium caseinate.

Means, in each column, followed by the same letter are not significantly different ($P \leq 0.05$) from each other according to DMR test.

Table 8
Influence of selected insecticide combinations on protease activity in black and red soils

Name of Pesticide	Soil incubation, in days			
	10 days	20 days	30 days	40 days
BLACK SOIL				
Control (0.0)	190 ^a	280 ^a	210 ^a	165 ^a
Profenofos + Cypermethrin (1.0 kg ha ⁻¹)	250 ^b	290 ^a	240 ^b	200 ^b
Deltamethrin + Endosulphan (2.5 kg ha ⁻¹)	270 ^c	310 ^b	260 ^c	215 ^c
RED SOIL				
Control (0.0)	165 ^a	210 ^a	185 ^a	145 ^a
Profenofos + Cypermethrin (1.0 kg ha ⁻¹)	230 ^b	260 ^b	196 ^b	115 ^a
Deltamethrin + Endosulphan (5.0 kg ha ⁻¹)	260 ^c	290 ^c	250 ^c	170 ^b

µg of tyrosine g⁻¹ soil formed after 24 h incubation with sodium caseinate.

Means, in each column, followed by the same letter are not significantly different ($P \leq 0.05$) from each other.

of the pesticides were either toxic or innocuous to soil microbial metabolic activities. Both black and red soil showed 6–85% invertase activity with pesticides alone, whereas a combination of pesticides showed 3–58% invertase activity. The insecticide combination

profenofos + cypermethrin has shown improved protease activity at 2.5 kg ha⁻¹ in both black and red soils. But with the combination of deltamethrin + endosulphan, there was variation in protease activity. An unexpectedly high concentration, i.e., 5.0 kg ha⁻¹, has shown maximum enzyme activity with nearly 57% high over control in red soil; in black soil, the enzyme activity was maximum at 2.5 kg ha⁻¹. Profenofos, deltamethrin, thiram, difenoconazole has shown high amounts of mineralisation of organic nitrogen at 2.5 to 5.0 kg ha⁻¹ in the black and red soil, whereas insecticide combinations, i.e., profenofos + cypermethrin and deltamethrin + endosulfan, have shown high amounts of mineralisation of organic nitrogen at 5.0 kg ha⁻¹ in both black and red soils. With pesticides alone, 1–21% of the ammonification, and 2–85% of nitrification were observed, whereas with a combination of pesticides showed 3–25% of the rate of ammonification and 23–40% of nitrification in both black and red soils. The results obtained in the present investigation clearly indicate that the application of pesticides alone and insecticide combination significantly enhanced the rate of sulphur oxidation specifically at 2.5 to 5.0 kg ha⁻¹ in both black and red soils. The effective combination of the stimulation of sulphur oxidation was profenofos + cypermethrin and Deltamethrin + endosulfan in the black and red soil. It is concluded that the application of pesticides (profenofos, deltamethrin, thiram, endosulphan) and insecticide combinations (profenofos + cypermethrin and Deltamethrin + endosulfan) at field application rates (2.5 to 5.0 kg ha⁻¹) improve the nitrogen mineralization, sulphur oxidation, protease and invertase activities in black and red soil.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.apsoil.2017.07.037>.

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